SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping

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speede3dgs.github.io

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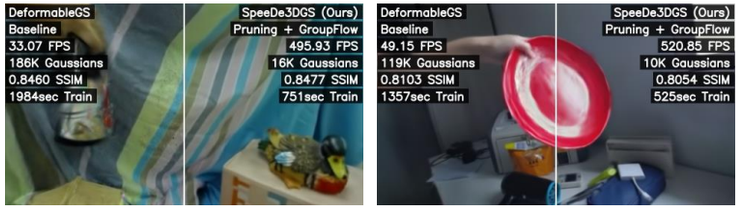
Background and Motivation

How can dynamic Gaussian Splatting deliver the high fidelity of neural motion while rendering 100+ FPS faster than non-neural motion models?



Dynamic Gaussian Splatting extends the set of 3D Gaussians $\mathcal{G}$ with a deformation network $\mathcal{D}$ that predicts time-varying offsets for the means μ, rotations r, and scales s at each timestep t:

$$(\mu + \Delta\mu_t,\ r + \Delta r_t,\ s + \Delta s_t) = \mathcal{D}(\mu, r, s, t).$$

$\mathcal{G}$ and $\mathcal{D}$ are jointly optimized over training frames $\mathcal{P}_{gt}$, where $I_{\mathcal{G}_t}(\phi)$ is the rendered image at pose ϕ and timestep t. g_i is the projected contribution of Gaussian $\mathcal{G}_i$ in $I_{\mathcal{G}_t}(\phi)$. Motion models for dynamic Gaussian Splatting methods fall in two categories:

- **Neural motion** is slow but produces high visual fidelity.
- **Non-neural motion** is fast but produces lower visual fidelity.

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Method

Temporal Sensitivity Pruning

We compute a **Temporal Sensitivity Pruning (TSP)** score $\tilde{U}_{\mathcal{G}_i}$ for each Gaussian $\mathcal{G}_i$ as the second-order sensitivity of the L_2 reconstruction error to its projected value:

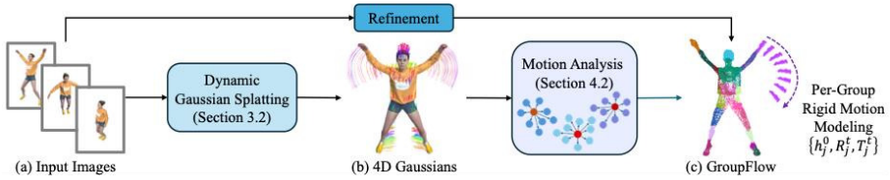
$$\tilde{U}_{\mathcal{G}_i} \approx \nabla^2_{g_i} L_2 \approx \sum_{\phi, t \in \mathcal{P}_{gt}} \left(\nabla_{g_i} I_{\mathcal{G}_t}(\phi)\right)^2.$$

Temporal Sensitivity Sampling (TSS) adds an annealing temporal perturbation to the timestamp input of the deformation network during TSP score computation, **revealing unstable primitives and suppressing floaters during pruning:**

$$(\mu + \Delta\mu, r + \Delta r, s + \Delta s) = \mathcal{D}(\mu, r, s, t + \mathcal{N}(0,1)\beta\Delta t(1 - i/\tau)).$$

We use our TSP + TSS score to prune low-sensitivity Gaussians during training, **reducing the total number of Gaussians by over 90% while preserving fidelity.**

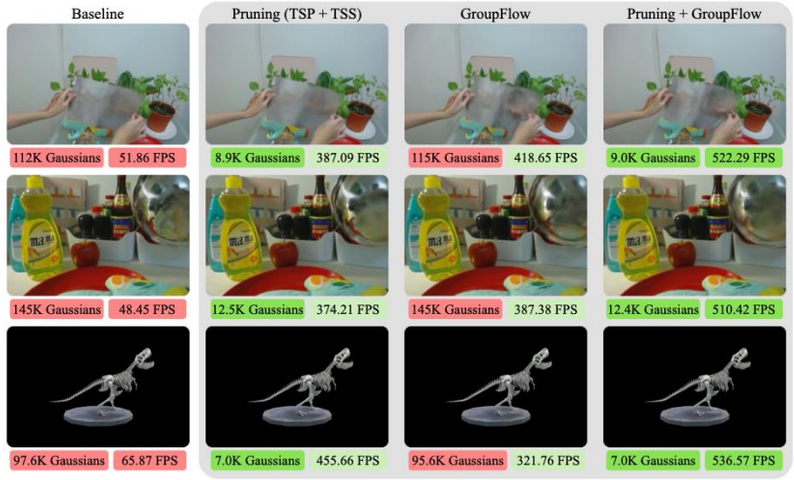
GroupFlow



Our GroupFlow method distills the high-fidelity per-Gaussian neural deformations into a smaller set of efficient grouped SE(3) transforms.

1. We start with a dense dynamic Gaussian Splatting model, where each Gaussian $\mathcal{G}_i$ is represented as a sequence of mean positions $\mathcal{M}_i = \{\mu_i^t\}_{t=0}^{F-1}$.
2. Then, we initialize J control trajectories $\{h_j^t\}$ and assign each mean μ_i to the most similar control point h_j using trajectory similarity score:
$$S_{i,j} = \lambda_r \operatorname{std}_t(\|\mu_i^t - h_j^t\|) + (1 - \lambda_r)\operatorname{mean}_t(\|\mu_i^t - h_j^t\|).$$
3. We fit a group-wise SE(3) flow to each group $\mathcal{M}^j$ using Umeyama alignment. The rotation R_j^t and translation T_j^t applied to $\mu_i \in \mathcal{M}^j$ at timestep t are:
$$\mu_i^t = R_j^t(\mu_i^0 - h_j^0) + h_j^0 + T_j^t.$$
4. The shared flow $\{h_j^0, R_j^t, T_j^t\}$ is optimized jointly with the scene, **reducing the number of transforms per frame from N (per-Gaussian) to J (per-group).**

Results



When compared to DeformableGS, **SpeeDe3DGS** achieves **11× FPS and 59% faster training on NeRF-DS** and **29× FPS and 73% faster training on HyperNeRF.**

Method	PSNR ↑	SSIM ↑	MS-SSIM ↑	LPIPS ↓	FPS ↑	Train Time (s) ↓
EffGS [20]	21.84	0.672	0.725	0.347	177.21	3757.81
STG-decoder [24]	21.81	0.678	0.742	0.352	109.42	5980.64
STG [24]	19.51	0.583	0.643	0.475	181.70	5359.56
RTGS [52]	21.61	0.663	0.720	0.350	143.37	7352.52
4DGS [47]	23.55	0.708	0.765	0.277	62.99 (1.00×)	8628.89 (1.00×)
+ Pruning (Ours)	22.44	0.689	0.737	0.334	**179.64 (2.85×)**	**4358.17* (1.47×)**
+ GroupFlow (Ours)	21.00	0.667	0.705	0.380	**290.21 (4.61×)**	**4176.49* (2.07×)**
DeformableGS [51]	24.07	0.694	0.755	0.283	20.20 (1.00×)	6227.43 (1.00×)
+ Pruning (Ours)	23.86	0.694	0.749	0.295	**137.01 (6.78×)**	**2850.60* (2.18×)**
+ GroupFlow (Ours)	23.52	**0.709**	**0.771**	0.313	**276.91 (13.71×)**	**2461.14* (2.53×)**

On the 50 scenes in MonoDyGauBench, **SpeeDe3DGS** achieves **100+ FPS higher speed and better visual fidelity** than all non-neural motion baselines.